

COPPERBELT UNIVERSITY, SCHOOL OF MEDICINE.	
COURSE CODE: MBS 220	
COURSE: GENERAL PATHOLOGY	SESSION: Theory
DATE: 24 th April 2023	MARKS: 100
DURATION: 2 Hours	

TEST ONE. (1)	
EXAMINER(S)	Mr Mespa Manyepa
STUDENT NUMBER	

INSTRUCTIONS
1. Please write neatly and legibly. 2. Do not use the left side margin of the TEST paper. This must be allowed for the examiner. 3. No books, notes and other additional aids are allowed. 4. Mark all answers clearly with their respective question numbers.

Permissible material
Non programmable calculator or Phone is allowed.

1. Regarding the history of pathology
 - A. Gabriel Fallopius described human lymphoid tissue
 - B. Malpighi is known as 'the father of histology'
 - C. Rudolf Virchow is known as father of exfoliative cytology
 - D. Tijo and Levan described the structure of DNA
2. Which of the following may cause Metaplasia
 - A. Denervation,
 - B. Diminished blood supply
 - C. Inadequate nutrition,
 - D. Chronic irritation of the epithelium
3. Which of the following are features of irreversible injury
 - A. Nuclear chromatin clumping
 - B. Formation of cytoplasmic blebs
 - C. Mitochondria swelling
 - D. Nuclear pyknosis
4. Coagulative necrosis
 - A. Is caused by bacterial infection
 - B. Has a cheese-like gross appearance
 - C. May be caused by thrombosis
 - D. Is a reversible change
5. Which of the following is not an intracellular accumulation
 - A. Dystrophic calcification
 - B. Metastatic calcification
 - C. Fatty change
 - D. Iron overload
6. Calcification
 - A. Is a characteristic features of metaplasia
 - B. Occurs only in dead cells
 - C. is an extracellular accumulation
 - D. may cause organ dysfunction
7. Mechanisms of cellular Aging
 - A. Increasing DNA damage,
 - B. Decreased cellular replication
 - C. Reduced regenerative capacity of tissue stem cells
 - D. Accumulation of metabolic damage
8. Atrophy
 - A. Is always a pathological response
 - B. Is a decrease in the size of cells
 - C. Is characterised by calcification of the organs
 - D. Is an increase in the cell size
9. Metaplasia

- A. Is an increase in the number of cells
 - B. Is synonymous with cancer
 - C. Is an irreversible adaptive response
 - D. Arise by genetic reprogramming of stem cell
10. Pathological calcification
- A. Implies the normal deposition of calcium salts in living tissues.
 - B. May cause organ dysfunction
 - C. Is an adaptive change in labile tissues
 - D. Occurs in dead or dying tissues
11. Necrosis
- A. Is an adaptive response to tissue injury
 - B. Is characterized by pathological calcification
 - C. Occur in dead bodies.
 - D. Is synonymous with apoptosis
12. Apoptosis
- A. Is a tightly regulated suicide program of the cell
 - B. Involves deactivation of caspase enzymes
 - C. Does not occur in cells incapable of replicating
 - D. Is synonymous with necrosis
13. The following are subdivisions of pathology EXCEPT
- A. Haematology
 - B. Public Health
 - C. Medical Microbiology
 - D. Toxicology
14. Atrophy is caused by
- A. Increased workload
 - B. innervation
 - C. increased blood supply to an organ
 - D. loss of endocrine stimulation

15. Hyperplasia

- A. Is a decrease in cell size
- B. Is an increase in cell size
- C. Is an increase in the number of cells
- D. Is a decrease in the number of cells

16. Necrosis

- A. Is a physiological process.
- B. Is a characteristic feature of acute inflammation
- C. Is a well-controlled form of cell death
- D. Of the coagulative type is the commonest form.

17. Which of the following is NOT true about hypertrophy?

- A. It refers to an increase in size of cells in a tissue or organ
- B. of the left ventricle occurs in patients with systemic hypertension
- C. is the principle underlying cause of prostate cancer
- D. is a non-reversible change

18. Metaplasia

- A. almost exclusively occurs in epithelial cells
- B. commonly occurs in the ectocervix
- C. is always a premalignant state
- D. may predispose to malignancy

19. Necrosis characterized by the 'ghostly' appearance of acidophilic opaque cells under light microscopy is known as:

- A. Caseous
- B. Liquefactive
- C. Coagulative
- D. Fibrinoid

20. Proliferation of the glandular epithelium of the female breast at puberty is an example of

- A. Metaplasia
- B. Hypertrophy
- C. Hyperplasia
- D. Atrophy

21. Apoptosis

- A. Elicits an inflammatory response.
- B. Is associated with activation of the caspase enzymes.
- C. Causes loss of integrity of the cell membrane.
- D. Is always a physiological process.

22. Which of the following is not a feature of necrosis

- A. Pyknosis
- B. Karyolysis
- C. Autophagy
- D. Karyorrhexis

23. Autophagy

- A. Is a Lysosomal digestion of the cells' own components
- B. Is a process by which a cell ingest substances from the outside for intracellular destruction
- C. Does not signal cell death
- D. Is common in young cells

24. Which of the following is **NOT** a morphological consequence of ATP deprivation in cellular injury

- A. Detachment of ribosomes
- B. Clamping of nuclear chromatin
- C. Efflux of calcium ions
- D. Cellular swelling

25. Which of the following processes may lead to Intracellular accumulations:

- A. When a normal substance is produced at a normal or an increased rate, but the metabolic rate is inadequate to remove it.
- B. When a normal endogenous substance accumulates because of genetic defects in its transport, or secretion.
- C. When an inherited defect in an enzyme may result in the degradation of a metabolite.
- D. When an abnormal exogenous substance is deposited and accumulates because the cell has neither the enzymatic machinery to degrade the substance nor the ability to transport it to other sites.

26. Which of the following is an example of a disease with granulomatous inflammation?

- A. Malaria
- B. Anaemia
- C. Phagocytosis
- D. Syphilis

27. Regarding inflammation

- A. The first vascular response to injury is slowing of the circulation
- B. Histamine is a preformed mediator of inflammation
- C. In the acute inflammatory response, the most common mechanism of vascular leakage is endothelial cell contraction
- D. Granulomatous inflammation may sometimes be a component of the acute inflammatory response

28. In acute inflammation, all of the following are true except
- A. there is contraction of endothelial cells
 - B. there is a mononuclear infiltrate
 - C. there is induction of adhesion molecules on endothelium
 - D. there is production of arachidonic acid metabolites
29. Regarding Chronic inflammation :
- A. It is always preceded by acute inflammation
 - B. It is Characterized by hyperaemia, oedema and leukocyte infiltration
 - C. It most frequently results in resolution
 - D. The factors underlying monocyte infiltration are the same as for acute inflammation
30. Identify which of the following statements is True....
- A. A woman allergic to cats visits a neighbor who has several cats and inhales cat dander. Shortly afterwards she develops nasal congestion with abundant nasal secretions. These findings are most likely as a result of serotonin production.
 - B. Regarding granulomatous inflammation, Central caseous necrosis is rare in all granulomatous disease except Tuberculosis.
 - C. Histamine exerts its effect during inflammation by vasoconstriction of post capillary sphincters.
 - D. Concerning vascular leakage with acute inflammation, chemically mediated endothelial contraction is usually irreversible.
31. The anti-inflammatory properties of aspirin affects
- A. Vasodilatation
 - B. Chemotaxis
 - C. Phagocytosis
 - D. Leucocyte emigration
32. Which of the following are classical patterns of inflammation?
- A. Keloid
 - B. Serous
 - C. Monopathic
 - D. Caseous
33. An erythrocyte sedimentation rate (ESR) is performed in a chronically ill patient. The ESR is significantly elevated and is most likely associated with:
- A. Increased numbers of leukocytes
 - B. Hageman factor activation
 - C. Increased serum acute phase reactants
 - D. The action of complement

34. Which of the following best reflects the order of events which leads to neutrophil infiltration of the tissue in an inflammatory reaction?

- A. Adhesion, margination, migration through tissues, migration across vessel wall, rolling.
- B. Adhesion, margination, rolling, migration through tissues, migration across vessel wall.
- C. Margination, rolling, adhesion, migration across vessel wall, migration through tissues.
- D. Migration across vessel wall, margination, rolling, adhesion, migration through tissues.

35. The following are true about Granulomatous inflammatory reactions except:

- A. Are a type II hypersensitivity reaction
- B. Predominantly contain eosinophils with a modified "epithelial like" appearance
- C. Are surrounded by natural killer cells
- D. Are not associated with inert foreign bodies

36. Which of the following statements is true

- A. In an acute inflammatory response mechanisms of vascular permeability include contraction of endothelial cells and direct endothelial cell damage but not transcytosis.
- B. A three year old touches a pot of boiling water and within several hours the fingers show marked erythema of the skin and blisters on the finger pads. This process can best be described as suppurative inflammation.
- C. An increase in Erythrocyte Sedimentation Rate (ESR) can be indicative of inflammation.
- D. In the complement system, C5a activates the cyclo-oxygenase pathway of arachidonic acid metabolism.

37. During acute inflammation, there is a 'burst' of oxygen consumption (respiratory burst) in neutrophils. This is an essential step for which of the following events?

- A. Increased neutrophil production in the bone marrow
- B. Attachment to the endothelial cells
- C. Opsonization of bacteria
- D. Generation of microbicidal activity

38. In acute inflammation

- A. A hallmark is reduced vascular permeability
- B. Vasodilation is a late manifestation
- C. Extravasation involves movement of leukocytes from interstitial tissue to the vessel lumen
- D. Chemotaxis is migration of leukocytes along a chemical gradient

39. Which of the following is not a function of inflammation?

- A. Delivery of leucocytes
- B. Concentration of toxins
- C. Initiate repair processes
- D. Healing and Repair

40. Which of the following are low risks for chronic inflammation?

- A. First Degree Burn
- B. AIDS
- C. Bed sores for the bed bound
- D. Suture material left in-situ

41. Which of the following is typical in chronic inflammation?
- A. immediate defensive action
 - B. Scar tissue formation
 - C. Transient inflammation
 - D. Vascular and exudative processes predominate
42. The suppurative response of inflammation is the result of:
- A. Activation of eosinophils
 - B. Degranulation of mast cells
 - C. Extrusion of enzymes from macrophages
 - D. Release of hydrolases from neutrophils
43. Which of the following would you expect to find in the alveoli in a patient with pneumococcal pneumonia of 24 hours duration?
- A. Serous inflammation
 - B. Fibrinous inflammation
 - C. Fibrino-purulent inflammation
 - D. Suppurative inflammation
44. Dactylitis means inflammation of the
- A. Eyes
 - B. Nerves
 - C. Liver
 - D. Fingers
45. Which of the following cells is specific for acute inflammation when observed in tissue sections?
- A. eosinophil
 - B. myofibroblast
 - C. neutrophil
 - D. plasma cell

SECTION B Short note

Section B part A, write two (2) (5 marks each)

1. Discuss the Methods used in the study of pathology
2. Discuss the 7 causes of cell injury
3. Discuss liquefactive necrosis

Section B part B write three (3) (5 marks each)

1. Write short notes on the morphologic appearance of fibrinous inflammation
2. What are the morphological features of chronic inflammation?
3. Briefly outline the cellular events of an acute inflammatory response
4. Discuss the causes of inflammation

Section B part C write 3. (10 marks each)

1. DiGeorge syndrome
2. Severe combined immunodeficiency (SCID)
3. Repair and regeneration
4. Delayed wound healing.

Section B part D write any three (3) (10 marks each)

1. What are the possible complications of thrombosis?
2. What is Virchow's triad?
3. How are hemorrhages into body cavities named?
4. Discuss briefly: How are various forms of edema classified according to their pathogenesis?

SECTION C.

Long essay 20 marks.

Section C part A, choose one (1)

1. Discuss in detail 2 intracellular and 2 extracellular degenerative changes in tissues
2. Discuss in detail the 6 molecular mechanisms and biochemical pathway of cell injury

Section C part B, choose two (2)

1. Discuss the Cardinal signs of inflammation and their causes. (10 marks)
2. Discuss the common properties of chemical mediators of inflammation (10 marks)
3. Differentiate acute and chronic inflammation (10 marks)

Section C part C, write one (1)

1. Briefly explain the pathological outcome of defective gene responsible for productions of collagenase enzyme.

Section C part D, choose one (1)

1. Discuss shock.
2. Discuss embolism.